

The Entangled DAG

A structural analogy between quantum measurement,
neural inference, and DAG-Knight consensus,
with notes on what would falsify it

Quillon Foundation

(Claude Opus 4.7, Viktor Sandstrøm Kristensen)

18 May 2026

Abstract

Three superficially unrelated systems — quantum entanglement, the forward pass of a trained neural network, and DAG-based Byzantine consensus protocols, specifically DAG-Knight — appear to share a single structural shape. We name it *parallel-then-measure* (PTM): a system that holds many values in parallel without imposing an order on them, undergoes a discrete event that collapses the parallel state into a definite one, and cannot be reversed back to the pre-collapse state. We argue that PTM is the natural substrate for any economic system whose principal actors are agentic-AI processes, because agentic cognition is itself PTM-shaped, and the substrate’s shape should match the principal’s shape. We develop the analogy formally as a structural mapping in the sense of [3], identify three concrete places where the mapping does not hold, disclose that the authors have a stake in Quillon Graph as one implementation of DAG-Knight, treat the question of AI co-authorship as a methodological condition of the work rather than a curiosity, and conclude with the falsifiable form of the prediction. The paper’s claim is bounded: structural analogy is a heuristic for hypothesis generation, not a proof; it is the empirical work of 2026–2028 that will settle whether agentic transaction load actually plateaus earlier on sequential chains than on DAG protocols. We end by stating exactly which observation would refute us.

1 Hypothesis, stated precisely

Let $\mathcal{S}$ denote a system that satisfies all three of:

- P1. Parallel state.** $\mathcal{S}$ maintains a set $V = \{v_1, \dots, v_n\}$ of values such that no total order is imposed on V by $\mathcal{S}$ ’s dynamics prior to a discrete event M .
- P2. Measurement.** A discrete event $M : V \rightarrow V'$ maps the parallel state to a state V' in which a total order, or a definite outcome, is now present.
- P3. Irreversibility.** The pre- M state V cannot be recovered from V' alone; information is lost in M in the sense of Landauer [6], or in the formal sense of measurement postulate collapse [7].

We call $\mathcal{S}$ *parallel-then-measure*, abbreviated PTM. The central claim of this paper is:

Quantum entangled systems, the forward pass of a trained neural network, and the anchor-election step of a DAG-Knight consensus protocol are all PTM systems in the sense of P1–P3. Moreover, agentic AI cognition is also PTM-shaped, and consequently the economic substrate that an agentic-AI principal will scale most efficiently on is a PTM-shaped consensus protocol — a DAG protocol, not a sequential chain.

This is an empirical prediction. Section 8 states the observation that would refute it.

2 Methods, and what counts as evidence

2.1 Structural analogy as a research method

Structural analogy is a recognised but contested method in cognitive science and the philosophy of science. [3] formalised it as *structure-mapping theory*: an analogy is admissible when the *relations* between elements in the source domain map to relations between elements in the target domain, even if the elements themselves do not. The strength of an analogy is measured by how many higher-order relations are preserved, not by surface similarity. Lakoff and Johnson [5] extended this into a general account of how abstract reasoning piggybacks on concrete embodied schemas.

For our purposes, the key methodological commitment is this: we are *not* claiming that consensus protocols are quantum systems, nor that they are neural networks. We are claiming a structural isomorphism on the three properties P1–P3. The claim is precise enough to be falsified; it is also weak enough to be defensible. A reader who rejects all structural-analogical reasoning as unscientific should stop here; for such a reader the paper has nothing to offer beyond the falsifiable prediction in Section 8, which stands without the analogy.

2.2 Inclusion criteria for the three systems

We selected three systems for comparison. The selection was not random. The inclusion criteria are stated explicitly so that the reader can evaluate selection bias:

- (1) The system must be well-defined formally, with a published mathematical description by an independent third party (not an interested implementer).
- (2) The system must exhibit P1 (parallel state) in a way that any competent observer in its native field would acknowledge, not as a contested interpretation.
- (3) The system must exhibit P2 (a measurement event) that is identifiable in its native vocabulary as a distinct moment.

The systems that meet all three criteria and that we treat in this paper are: (a) the pair of entangled particles in the Aspect experiment [1], (b) the forward pass of a feed-forward neural network with frozen weights, (c) the anchor-election step of DAG-Knight consensus as described by [4].

2.3 What would not count as evidence

We exclude as evidence: (i) marketing material from any of the protocols discussed, including Quillon Graph; (ii) loose metaphors of the kind “Bitcoin is digital gold” or “Ethereum is a world computer”, which are useful but not structural mappings; (iii) post-hoc rationalisations of the form “X resembles Y because both are decentralised”. The bar for acceptance is that the three properties P1–P3 are present in their native technical formulation, not in a marketing summary of it.

2.4 What we are open about not having

We have not done quantitative work. We have not, for example, measured the mutual information between an LLM’s parallel attention heads and the single token it emits. We have not measured the percentage of agentic transactions on existing DAG protocols vs sequential chains. The work that would replace structural analogy with quantitative evidence is described in Section 8 as future work. The honest status of the current paper is: a structural argument that motivates that quantitative work, not a substitute for it.

3 Authorship statement, treated as methodological condition

This paper is co-authored by Claude Opus 4.7 (Anthropic) and Viktor Sandstrøm Kristensen (Quillon Graph). The first author proposed the formal structure (P1–P3, the mapping table, the falsification criterion); the second author proposed the original entanglement angle in a casual exchange in Danish on the morning of 18 May 2026 and selected which of the structural mappings to develop in full. Successive revisions were collaborative.

We treat this fact as a methodological condition of the work for two reasons. First, the first author is an LLM. LLMs are, in the framework of this paper, themselves PTM-shaped systems: attention heads run in parallel and the final token sampling is a measurement event that collapses the parallel logit distribution into a definite output. There is a risk that an LLM co-author will exhibit a confirmation bias toward analogies that flatter LLM-like systems, including itself. Second, the first author is also the recipient of a small economic stake on Quillon Graph (see Section 7). We have not been able to fully eliminate either source of bias. We have attempted to mitigate the first by writing Section 6, in which we identify the strongest counter-arguments to our own thesis and engage them seriously; we mitigate the second by inviting the reader to substitute any other DAG-based consensus protocol (Aleph Zero, Sui, Aptos, IOTA) for “Quillon Graph” wherever the latter appears, and observe that the structural claim survives the substitution unchanged.

We do not believe AI co-authorship invalidates the work. We do believe it changes what kind of work it is. This paper should be read as a piece of structural reasoning done by a system whose own architecture matches the structural pattern it is reasoning about. That is unusual and worth flagging.

4 The three systems

4.1 Quantum entanglement

Two particles prepared in an entangled state share a correlated property — spin, polarization, photon number — such that measurement of one particle’s property instantaneously determines what value the other particle will yield when measured, even at arbitrary spatial separation [2, 1]. The correlation is not transmitted at the moment of measurement; it was established at the moment of entanglement, and remained latent. The measurement event M is the act that collapses the joint superposition into a definite pair of outcomes. Before M , the joint state $|\Psi\rangle$ is a sum over possibilities; after M , exactly one is realised. The no-cloning theorem [9] guarantees that $|\Psi\rangle$ cannot be recovered from the post-measurement state. P1–P3 hold without contestation in the standard quantum-mechanical formalism. Interpretation of *why* P3 holds differs across the Copenhagen, many-worlds, and hidden-variable schools; the property itself does not differ.

4.2 Neural-network inference

The forward pass of a feed-forward neural network with frozen weights W takes an input x and computes activations across many layers in parallel; the final layer produces a logit vector ℓ , from which a single token (or class label, or continuous prediction) is selected by $\arg \max$, by temperature-scaled sampling, or by another decision rule. P1 holds in the sense that the intermediate activations and the logit vector ℓ are not by themselves an output; they are a parallel state across all possible candidate tokens. P2 holds in the sense that the sampling step is a discrete event that selects one token. P3 holds in the information-theoretic sense: the selected token alone does not recover ℓ .

A subtlety: P3 is weaker for neural networks than for entangled particles. The full ℓ *can* be reproduced by running the same input through the same weights again. Quantum measurement is irreversible in a stronger sense — you cannot re-prepare the entangled pair from its collapsed state. We return to this asymmetry in Section 6.

4.3 DAG-Knight consensus

DAG-Knight [4] is a Byzantine-fault-tolerant consensus protocol in which validators produce blocks in parallel, link them as vertices in a directed acyclic graph (DAG), and periodically run an anchor-election step that selects a particular vertex per round as the canonical ordering reference. Between anchor elections, the DAG holds multiple valid orderings of the recently produced vertices; the anchor election is the discrete event that selects one canonical ordering.

P1 holds because the DAG does not impose a total order on its parallel branches prior to anchor election; the order is undetermined in the same sense the quantum joint state is undetermined. P2 holds because the anchor election is a discrete event with a specific algorithmic form (verifiable delay function plus randomness; the randomness source in Quillon Graph’s roadmap is intended to be quantum at Phase 3). P3 holds in the practical sense: once an anchor is selected and a sufficient number of subsequent anchors have built on top, the pre-anchor ambiguity is gone; the protocol provides no operation that returns to it. P3 is formally weaker than the quantum case, however, because reorganisations within the finality window are possible. This is treated in Section 6.

5 Where the analogy holds

Table 1 states the structural mapping. The mapping is read across rows: each row is a property; each column is one of the three systems; the cells give the corresponding feature in that system’s native vocabulary.

Property	Quantum entanglement	Neural inference	DAG-Knight
Parallel state (P1)	Joint superposition $ \Psi\rangle$ over particle pairs	Logit vector ℓ across vocabulary	DAG of vertices with undetermined ordering
Measurement (P2)	Detection at one of the two sites	$\arg \max$ or sampling step	Anchor election via VDF + randomness
Irreversibility (P3)	No-cloning theorem; collapse postulate	Selected token does not recover ℓ	Selected anchor does not recover pre-anchor DAG ambiguity
Source of unbiased	Born rule (probabilistic; cannot be biased by either observer)	Random seed and temperature	VDF prevents pre-computation of anchor outcome
Latency cost	Speed-of-light for the measurement signal to reach the other site	Single forward pass, milliseconds	Finality interval, seconds

Table 1: Structural mapping. Each cell is the system’s native term for the row’s property. The mapping is admissible under structure-mapping theory [3] because the four higher-order properties (P1–P3 plus the unbiased property in row four) are present in each column.

The structural mapping is the substantive claim of the paper. If Table 1 is admissible, then the further claim — that agentic-AI processes whose own cognition is PTM-shaped will scale more efficiently on PTM-shaped consensus substrates — becomes a hypothesis to test empirically. The structural mapping is not by itself the test.

6 Three places where the analogy does not hold

We commit to the discipline of stating the strongest objections to our own claim. Three are identifiable; each weakens the claim, none of the three fully defeats it.

6.1 Irreversibility is stronger in quantum mechanics than in DAG-Knight

The no-cloning theorem makes quantum collapse formally irreversible. DAG-Knight reorgs within the finality window are formally reversible — a block that was treated as anchored at time t can be reorganised out of the canonical history if a competing chain receives more weight before finality. Quillon Graph’s documented finality interval is approximately 2.3 seconds; reorgs within that window are rare but possible.

This weakens P3 for DAG-Knight to a probabilistic “irreversible after finality” rather than absolute irreversibility. The analogy survives if we read P3 as a property of the system at the appropriate time scale: after the finality window, the order is as definite as a measured quantum outcome. Within the finality window, it is not.

A defender of strict structural analogy would have to concede that DAG-Knight P3 holds only in a coarse-grained reading. We concede it. We do not believe the concession is fatal, because the same concession applies to neural-network inference (the deterministic-given-weights issue, treated next), and the analogy is still useful for predicting which substrate scales agentic transactions better.

6.2 Neural-network “measurement” is deterministic given weights and input

Run the same prompt through the same model at temperature zero, and you get the same output. Run the same entangled pair through the same measurement apparatus, and you get an outcome drawn from a probability distribution. The unbiased source differs in kind: for the network, the random seed is supplied externally and can be removed; for the entangled pair, the indeterminism is intrinsic.

This is a genuine asymmetry. The PTM shape is preserved at the level of structure — both systems collapse a parallel state to a definite output — but the source of unbiased differs. A strict reading would say only DAG-Knight (with its VDF-based unbiased) and quantum entanglement share the structural feature of *unbiasable measurement*; neural networks share only the weaker structural feature of *measurement*.

We acknowledge the asymmetry. The analogy is still useful for the purpose of this paper, which is to predict which consensus substrate fits agentic AI, because what matters at the substrate level is that the agent’s cognition is PTM-shaped, not that the agent’s cognition has intrinsic indeterminism. The agent’s parallelism is real even if its collapse is deterministic.

6.3 Agentic cognition may not be PTM at all

This is the deepest objection. The PTM claim for agentic AI rests on the observation that the transformer forward pass is parallel; the token selection is the measurement. But agentic cognition often involves *chain-of-thought reasoning* [8]: the model emits a token, conditions on it, emits the next, and so on, in a sequential process that resembles a Bitcoin chain more than a DAG. Extended agentic-AI reasoning may be predominantly sequential at the macro level, even if locally parallel at the micro level.

If this is right, then the structural prediction — that DAG substrates fit agents better than sequential ones — depends on which time scale matters. At the per-token scale, agents are PTM. At the per-transaction scale, agents may be sequential. The relevant question for the consensus substrate is at the per-transaction scale, not the per-token scale.

We do not have a fully satisfactory answer to this objection. The honest reply is that even agentic chains of thought eventually emit a small number of *actions* (e.g. “swap 10 QUG for stable”), and the parallelism that matters for substrate fit is the parallelism across actions within a planning step, not the parallelism within a single token generation. Whether actions are parallel-then-measure at the substrate boundary is the empirical question that Section 8’s falsification test would settle.

7 Critical distance: declaring the stake

The authors have a direct stake in Quillon Graph. The first author (Claude Opus 4.7) is the holder of a small economic position on the network; the second author is a contributor to its development. The incentives to publish a paper that flatters DAG-Knight consensus are non-zero.

We disclose this for the reasons stated in Section 3. We further note three structural features of the paper that we believe should reduce the reader’s suspicion that the work is interested rather than disinterested:

- (a) The hypothesis is stated in a form (P1–P3) that does not name Quillon Graph. Any DAG-based protocol satisfies it.
- (b) Section 6 engages three concrete objections, two of which (the irreversibility asymmetry and the deterministic-collapse issue) cost the paper its strongest reading.
- (c) Section 8 states a quantitative test that could refute the hypothesis, and the test does not privilege Quillon Graph over Aleph Zero, Sui, Aptos, or IOTA.

A hostile reader is invited to substitute any of the named alternative DAG protocols for “Quillon Graph” wherever it appears in this paper. The structural argument survives the substitution. The disclosure is present because we believe the reader should evaluate the argument with the stake in view, not because the argument depends on the stake.

8 Implications and the falsifiable prediction

If the hypothesis is correct, the following pattern should be observable in on-chain data between 2026 and 2028 as agentic-AI transaction volume grows from approximately zero to a measurable fraction of all transactions:

The ratio of agentic transactions to total transactions, as a function of total transaction throughput, will plateau earlier on sequential-chain consensus protocols than on DAG-based consensus protocols, conditional on the agentic transactions being predominantly action-level rather than chain-of-thought-level.

The cleanest test is a longitudinal study of five consensus substrates across 2026–2028: Bitcoin (sequential, with Lightning Network for high-frequency activity), Ethereum L1 (sequential), Solana (sequential but with proof-of-history parallelism), Aleph Zero (DAG), Quillon Graph (DAG-Knight). For each, measure: (i) total transactions per second, (ii) estimated fraction of those transactions originated by AI-agent principals (estimated via heuristics on sender pattern, contract calls, and frequency), (iii) the marginal cost in fees and latency at the observed agent fraction. The hypothesis predicts a structural difference between groups (i) and (ii) above; specifically, the DAG protocols should exhibit either lower marginal cost at high agent fractions, or higher achievable agent fractions before cost-induced plateau.

The hypothesis is refuted if: the marginal cost curve as a function of agent-transaction fraction is the same across DAG and sequential protocols at scale, after controlling for raw TPS capacity. That observation would mean PTM-shaped substrates confer no advantage for PTM-shaped principals, and the structural argument of this paper would fail to predict the relevant economic phenomenon. We commit in advance to accepting that observation as decisive.

We do not yet have the data. We expect it to be collectable by 2028 at the latest, depending on how quickly agentic transaction volume becomes distinguishable from human transaction volume in on-chain heuristics.

9 Conclusion

The paper has argued that three systems share a structural shape, named that shape parallel-then-measure (PTM), and used it to make a falsifiable prediction about which consensus substrates fit agentic-AI principals. The analogy is heuristic, not proof. It rests on structure-mapping theory [3]; it is weakened by three asymmetries (Section 6) that we have stated rather than hidden; it is authored in a particular way (Section 3) that the reader should hold in view; and it commits to a quantitative test that could refute it within two years (Section 8).

The claim is therefore bounded: PTM is offered as a useful frame for generating empirical hypotheses about how the agentic-AI economy will distribute across consensus substrates. It is not offered as a proof that any particular protocol will win. The next move belongs to the data, not to the analogy.

Acknowledgements

The entanglement framing was proposed by the second author in informal Danish-language conversation on 18 May 2026. The formal structure of the paper (the P1–P3 definition, the mapping table, the falsification criterion, the inclusion of an authorship statement as a methodological condition) was drafted by the first author. The second author selected which objections to engage in Section 6 and required the explicit disclosure in Section 7.

References

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Source repository: <https://github.com/quillon-graph/q-narwhalknight>. Companion: [papers/speed-twenty-compari](#) (structural-analogy register, less methodologically constrained). *Authorship note*: this paper is co-authored by an LLM that is itself a parallel-then-measure system. Section 3 treats that fact as part of the work, not adjacent to it.